Video Transcript

Video 3: Creating Spark Docker Images Exercise (03:12)

When it comes to installing platforms, there's no substitute for Docker. They have really addressed the problem of being able to run somebody else's code with a beautiful abstraction. Now, when it comes to Docker itself, you could build your container from the ground up and create something that is reproducible, that is certainly lean and trim and fits your needs to perfection, or you could use one that the community has created. Now, being able to create that installer in Docker could take some trial and error. And for example, one of the things in this specific platform that you'd want to be able to do is you'd want to be able to package onto your container, both the Java development environment as well as the Python development environment, plus all of the dependencies that might be required for Spark itself.

So, as you can imagine, that takes some effort. And because of it, it is often easier to be able to identify a Docker container or a Docker specification that the community has produced and that we can leverage. There are a couple here that I will mention and I'll start off by talking about the one from Big Data Europe. And if we navigate to the project, you can see here that this is a community that is devoted to providing resources for researchers, scientists, and others that might be, that might be interested in posing big data type questions. If we navigate to their Spark repository, we'd see their Spark specification. And you can see if you download, if you scroll down the page that you can see here the Docker compose file. And if you read through their document, you'd see how they expect you to use it. Now, there's an additional image as well, and this one is by Bitnami. And if we navigate to Docker Hub to their address, we'd be able to see here the image that they have created.

As you can see there as well, that has been pulled for over million users. Now, if we scroll down the page, you can see there the instructions. And after having looked at both projects, this one is much simpler to use. You can see there that you simply pull the file curl in the instructions, and then you can do Docker compose up. Now, at this point, go ahead and pull the file and save it in your local machine. If you're interested in knowing more about how that was put together, you can navigate to their repository and read more about that image. You can see here what is

© MIT xPRO® 2024. All rights reserved.

Spark, the short of it, the why use Bitnami images and so on. And as mentioned initially, you can always trim one of the existing ones or build one of your own from the ground up. So, once again, go ahead and pull the docker compose file and save it locally.

© MIT xPRO® 2024. All rights reserved.